

EURO MOTORS

(Private) Limited

IT INVENTORY MANAGEMENT SYSTEM

System Documentation

Prepared by: IT Department

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1. Introduction

This document provides a comprehensive overview of the IT Inventory Management System developed for Euro Motors (Private) Limited. It covers the business context, system objectives, architecture, functional modules, security model, data design, deployment strategy and future roadmap.

The system replaces the manual spreadsheet-based inventory tracking previously used by the IT Department and introduces a centralised, database-driven web application for managing all company IT assets throughout their lifecycle.

2. Company Profile

Euro Motors (Private) Limited operates across multiple divisions and branch locations. The company's operational structure includes the following principal units:

Division / Department
Head Office
Passenger Vehicle Division
Commercial Vehicle Division
Bus Division
SKD and Assembly Operations
Service and Spare Parts
Branch Network
Sales Departments
Administration
Finance
Human Resources
IT Department

The system is designed to be extensible, allowing new branches, departments, asset categories and employees to be added as the organisation grows.

3. Business Objectives

The IT Inventory Management System is intended to achieve the following business objectives:

#	Objective
1	Maintain a complete register of all IT assets
2	Identify the current location of each asset
3	Identify the employee currently using each asset
4	Track asset assignment and return history
5	Track repairs, maintenance and warranty details
6	Monitor available, assigned, damaged, repaired and retired assets
7	Reduce duplicate or missing inventory records
8	Improve accountability for company-owned IT equipment
9	Produce management and audit reports
10	Restrict data modification to authorised IT staff
11	Generate and scan QR codes for physical asset verification
12	Maintain an audit trail of system activities

4. System Architecture

4.1 High-Level Architecture

The application follows a three-tier architecture consisting of a React single-page application on the front end, a Node.js REST API on the back end, and a PostgreSQL relational database for persistent storage. The tiers communicate exclusively through HTTP-based API calls secured with JSON Web Tokens (JWT).

4.2 Technology Stack

Layer	Technology	Details
Frontend	React + Vite	Single-page application with Tailwind CSS styling
Routing	React Router	Client-side route management with role-based guards
Backend	Node.js + Express.js	RESTful API server with middleware pipeline
ORM	Prisma 7	Type-safe database access with schema-driven migrations
Database	PostgreSQL 16+	Relational database running on port 5433
Authentication	JWT	Token-based stateless authentication
Database Admin	pgAdmin	GUI administration and query tool

4.3 Development Environment

Service	URL / Port
Backend API	http://localhost:5000/api
Frontend Dev Server	http://localhost:5174
PostgreSQL	Port 5433
Database Name	euromotors_it_inventory

5. System Modules

The system is organised into fourteen functional modules, each addressing a distinct area of IT inventory management:

#	Module	Purpose
1	Authentication	User login, JWT token issue and session management
2	Dashboard	Management overview with key metrics and alerts
3	User Management	Create, edit and deactivate system user accounts
4	Departments	Maintain the company department master list
5	Locations	Maintain the branch and site location master list
6	Employees	Employee records linked to departments and locations
7	Assets	Full lifecycle IT asset register
8	Asset Assignments	Assign assets to employees with history tracking
9	Asset Returns	Record asset returns and update assignment status
10	Maintenance & Repairs	Log repairs, costs, vendors and resolution
11	QR Code Management	Generate, print and scan QR labels for assets
12	Reports	Filtered, exportable management and audit reports
13	Audit Logs	Immutable record of all system activities
14	Settings	System configuration and administrative preferences

6. User Roles and Permissions

The system implements role-based access control (RBAC) with three defined roles. Permissions are enforced both in the frontend (UI element visibility) and the backend (API middleware validation).

6.1 Role Summary

Role	Access Level	Description
Admin	Full Access	Complete control over all modules, users, and settings
IT Officer	Operational Access	Can create and edit operational records but cannot manage users, delete master data, or access settings
Viewer	Read-Only Access	Can view permitted information across modules but cannot create, edit, or delete any records

6.2 Detailed Permission Matrix

Module / Action	Admin	IT Officer	Viewer
View Dashboard	Yes	Yes	Yes
View Departments	Yes	Yes	Yes
Add / Edit Departments	Yes	No	No
Delete Departments	Yes	No	No
View Locations	Yes	Yes	Yes
Add / Edit Locations	Yes	No	No
Delete Locations	Yes	No	No
View Employees	Yes	Yes	Yes
Add / Edit Employees	Yes	Yes	No
Delete Employees	Yes	No	No
View Assets	Yes	Yes	Yes
Add / Edit Assets	Yes	Yes	No
Delete Assets	Yes	No	No
Create Assignments	Yes	Yes	No
Record Returns	Yes	Yes	No
Manage Maintenance	Yes	Yes	No
Generate / Scan QR Codes	Yes	Yes	No
View Reports	Yes	Yes	View only
View Audit Logs	Yes	No	No
Manage Users	Yes	No	No
System Settings	Yes	No	No

7. Authentication and Security

7.1 Authentication Flow

The system uses JSON Web Token (JWT) authentication to secure all protected endpoints. The authentication process operates as follows:

Step	Action	Component
1	User submits login credentials (username and password)	Frontend login form
2	Backend validates credentials against the database	Auth controller
3	Backend generates and returns a signed JWT token	Auth service
4	Frontend stores the JWT token in browser local storage	React application
5	Token is attached to the Authorization header of all protected API requests	HTTP client
6	Backend middleware validates the token and extracts user role on every request	auth.middleware.js

7.2 Security Measures

- Passwords are hashed before storage; plain text passwords are never retained
 - JWT tokens carry an expiry time to limit session duration
 - Role-based middleware rejects requests from users without sufficient privileges
 - Frontend navigation elements are conditionally rendered based on user role
 - Backend permission checks are enforced independently of frontend restrictions
 - Environment variables store secrets; credentials are never committed to source control
-

8. Database Design

8.1 Database Configuration

Parameter	Value
Database Engine	PostgreSQL 16+
Database Name	euromotors_it_inventory
Port	5433
ORM	Prisma 7
Initial Migration	20260426081350_init_it_inventory
Config File	prisma.config.ts (Prisma 7 format)

8.2 Database Enumerations

The schema defines the following enumerated types to enforce data consistency across the application:

Enum Name	Values	Usage
AssetStatus	AVAILABLE, ASSIGNED, UNDER_REPAIR, RETIRED	Current state of an IT asset
AssignmentStatus	ACTIVE, RETURNED	Current state of an asset assignment
MaintenanceStatus	OPEN, IN_PROGRESS, RESOLVED	Current state of a maintenance request
GeneralStatus	ACTIVE, INACTIVE	General-purpose status for master records

8.3 Key Relationships

The database schema implements the following principal entity relationships:

- Employee to Assets – One-to-many direct assignment relationship using a named Prisma relation (`@relation("DirectAssignment")`)
- Asset to Employee – Each asset may reference one currently assigned employee
- Assignment records – Separate join entity preserving full assignment and return history
- Department to Employees – One-to-many; each employee belongs to one department
- Location to Employees – One-to-many; each employee is assigned to one location
- Asset to Maintenance – One-to-many; each asset may have multiple maintenance records

9. Asset Data Model

Each IT asset record captures the following data attributes across multiple categories:

9.1 Identification Fields

Field	Description	Constraint
Asset Code	Unique internal identifier (e.g. EM-IT-LAP-0001)	Unique, Required
Asset Name	Descriptive name of the asset	Required
Asset Category	Classification (Laptop, Desktop, Printer, etc.)	Required
Serial Number	Manufacturer serial number	Unique
Barcode	Barcode identifier if applicable	Optional
Brand	Manufacturer or brand name	Optional
Model	Product model designation	Optional

9.2 Purchase and Warranty Fields

Field	Description
Purchase Date	Date the asset was acquired
Purchase Price	Cost of acquisition
Supplier	Vendor or supplier name
Invoice Number	Purchase invoice reference
Warranty Start Date	Date warranty coverage begins
Warranty Expiry Date	Date warranty coverage ends

9.3 Allocation Fields

Field	Description
Assigned Employee	Employee currently using the asset
Department	Department the asset is allocated to
Location / Branch	Physical location of the asset
Assignment Date	Date the asset was assigned

9.4 Technical Specification Fields

Field	Applicable To
Processor	Laptops, Desktops, Servers
RAM	Laptops, Desktops, Servers

Field	Applicable To
Storage	Laptops, Desktops, Servers
Operating System	Laptops, Desktops, Servers
IP Address	Network equipment, Servers
MAC Address	Network-connected devices
Hostname	Networked devices
License Details	Software licenses

10. Asset Categories

The system supports the following initial asset categories. Additional categories can be added through the master-data management function without code changes:

Category	Category	Category
Laptop	Desktop Computer	Monitor
Printer	Scanner	Server
Network Switch	Router	Firewall
Access Point	UPS	Mobile Phone
Tablet	Projector	CCTV Equipment
Biometric Device	Keyboard	Mouse
External Storage Device	Software License	Other IT Equipment

11. Asset Assignment Module

The assignment module manages the allocation of IT assets to employees and maintains a complete historical record of all assignment transactions.

11.1 Assignment Data Fields

Field	Description
Asset	The IT asset being assigned
Employee	The receiving employee
Department	Employee's department at time of assignment
Location	Physical location of assignment
Assignment Date	Date the asset was handed over
Expected Return Date	Planned return date (if applicable)
Actual Return Date	Date the asset was physically returned
Assigned By	User who created the assignment
Returned To	User who processed the return
Condition at Assignment	Asset condition when issued
Condition at Return	Asset condition when received back
Assignment Status	ACTIVE or RETURNED
Notes	Additional remarks

11.2 Assignment Business Rules

When an asset is assigned:

- The asset status changes to ASSIGNED
- The assigned employee reference is updated on the asset record
- A new assignment history record is created with status ACTIVE

When an asset is returned:

- The assignment status changes to RETURNED
- The actual return date is recorded
- The asset status reverts to AVAILABLE (unless repair or retirement is needed)
- The assigned employee reference is cleared from the asset record
- The historical assignment record is preserved permanently

12. Maintenance and Repairs Module

The maintenance module logs all repair and service activities for IT assets, including internal repairs, external vendor services, warranty claims and associated costs.

12.1 Maintenance Data Fields

Field	Description
Asset	The asset requiring maintenance
Issue Reported	Description of the fault or problem
Reported Date	Date the issue was raised
Reported By	Person who reported the issue
Assigned Technician	Internal technician handling the repair
Vendor	External repair vendor (if applicable)
Maintenance Status	OPEN, IN_PROGRESS, or RESOLVED
Estimated Cost	Projected repair cost
Actual Cost	Final confirmed cost
Sent for Repair Date	Date sent to external vendor
Returned Date	Date received back from repair
Resolution	Description of the fix applied
Notes	Additional remarks

12.2 Maintenance Business Rules

When an asset is sent for repair, its status changes to UNDER_REPAIR. Upon resolution, the asset status is updated to AVAILABLE, ASSIGNED or RETIRED depending on the asset's actual condition and continued usability.

13. QR Code Management

Each IT asset will have a unique QR code linked to its database record. The QR code system supports the following operations:

- Generate a QR code containing the asset identifier or a secure lookup reference
- Print QR labels for physical attachment to assets
- Scan QR codes using a mobile device or barcode scanner
- Retrieve and display the full asset record from a scanned code
- Support physical stock verification exercises

QR codes do not expose sensitive information directly. The encoded data contains only the minimum reference needed to look up the full record via authenticated API access.

14. Reports

The system provides a comprehensive suite of management and operational reports:

Report Category	Reports
Asset Register	Complete asset register, assets by category, assets by brand
Allocation	Assets by department, assets by location, assets by employee, unassigned assets
Status	Available assets, assigned assets, under repair, retired assets
Assignment History	Asset assignment history, employee asset history
Maintenance	Maintenance history, maintenance cost report
Procurement	Assets purchased during a selected period
Warranty	Warranty-expiring assets
Verification	Missing or unverified assets
User Activity	User activity and audit report

All reports support filtering by date range, department, location, status and category where applicable. Export formats include PDF, Excel (XLSX), CSV and a browser-based print view.

15. Dashboard

The dashboard provides an at-a-glance management overview with the following key performance indicators:

Indicator	Description
Total Assets	Count of all registered IT assets
Available Assets	Assets not currently assigned or under repair
Assigned Assets	Assets currently allocated to employees
Under Repair	Assets currently in maintenance
Retired Assets	Assets decommissioned from active use
Total Employees	Count of registered employees
Total Departments	Count of active departments
Total Locations	Count of active locations
Warranty Alerts	Assets with warranties expiring soon
Open Maintenance	Unresolved maintenance requests
Recent Assignments	Latest asset allocation activity
Recent Additions	Newly registered assets

16. Audit Logs

The audit log module creates an immutable record of significant system events to support accountability, compliance and troubleshooting.

16.1 Tracked Events

Category	Events
Authentication	User login, failed login attempts
User Management	User creation, update, deletion
Assets	Asset creation, update, deletion
Assignments	Assignment creation, asset returns
Maintenance	Maintenance record creation, status updates
Master Data	Department, location and employee changes

16.2 Audit Record Fields

Field	Description
User	The user who performed the action
Action	The type of event (CREATE, UPDATE, DELETE, LOGIN, etc.)
Module	The system module affected
Record ID	The identifier of the affected record
Date and Time	Timestamp of the event
Old Values	Previous field values (for updates)
New Values	Updated field values (for updates)
IP Address	Client IP address (where feasible)

17. Backend API Structure

The Express.js backend exposes the following RESTful API route groups, each handling a distinct resource domain:

Route Group	Base Path	Responsibilities
Auth	/api/auth	Login, token generation, session validation
Users	/api/users	User account CRUD operations
Departments	/api/departments	Department master data management
Locations	/api/locations	Location and branch management
Employees	/api/employees	Employee record management
Assets	/api/assets	IT asset lifecycle management
Assignments	/api/assignments	Asset assignment and return processing
Maintenance	/api/maintenance	Repair and service record management

All routes use consistent error handling, input validation, role-based middleware and standardised JSON response formats. Protected endpoints require a valid JWT token in the Authorization header.

18. Development Status

The following table summarises the current completion status of each development phase and module:

Phase / Module	Status	Notes
Phase 3 – Backend & Database	Completed	PostgreSQL, Prisma, Express, JWT, routes
Module 1 – Login & Roles	Completed	All three roles tested and verified
Module 2 – Dashboard	Completed	Eight key metrics operational
Module 3 – Departments & Locations	Completed	CRUD with role-based access
Module 4 – Employees	Completed	Full CRUD, linked dropdowns, role-based
Module 5 – Assets	In Progress	Next priority for frontend completion
Assignments Module	Planned	Following Assets completion
Returns Module	Planned	Following Assignments
Maintenance Module	Planned	Following Returns
QR Code Module	Planned	Post-core module completion
Reports Module	Planned	Post-core module completion
Audit Logs	Planned	Post-core module completion
Deployment	Planned	Vercel (frontend) + Render (backend)

19. Deployment Strategy

19.1 Target Platforms

Component	Platform	Notes
Frontend	Vercel	Connected to GitHub repository for auto-deploy
Backend API	Render	Connected to GitHub repository for auto-deploy
Database	Managed PostgreSQL	Cloud-hosted PostgreSQL instance

19.2 Deployment Workflow

The deployment process follows a Git-based continuous deployment workflow:

- All changes are developed and tested on the local development environment
- Tested changes are committed to the Git repository
- Code is pushed to the designated GitHub branch
- Vercel automatically rebuilds and deploys the frontend
- Render automatically rebuilds and deploys the backend
- Database migrations are executed carefully in the production environment

19.3 Environment Configuration

Each deployment environment requires the following configuration:

- Frontend: API base URL pointing to the deployed backend (not localhost)
- Backend: Production database connection URL, JWT secret, allowed CORS origin, port
- All secrets are stored as environment variables and are never committed to source control

20. Future Enhancements

The following enhancements are planned for future development phases:

Category	Enhancement
Mobile	Mobile-responsive asset scanning and verification
QR Codes	QR label printing and batch generation
Notifications	Scheduled warranty expiry alerts and email notifications
Workflow	Approval workflows for asset transfers and disposals
Operations	Physical stock verification and asset transfer between branches
Finance	Depreciation tracking and procurement integration
Integration	SAP Business One integration
Authentication	Active Directory / Microsoft SSO login
Data Management	Excel bulk import and employee offboarding workflow
Documentation	Digital asset handover forms and electronic acknowledgement
Attachments	Photo, invoice and repair document attachments
Reporting	Advanced dashboard charts and audit analytics
Infrastructure	Automated backup and disaster recovery

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This document should be updated as modules are completed, new features are added, or system architecture changes occur. All amendments should be recorded in a revision history maintained alongside this document.

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